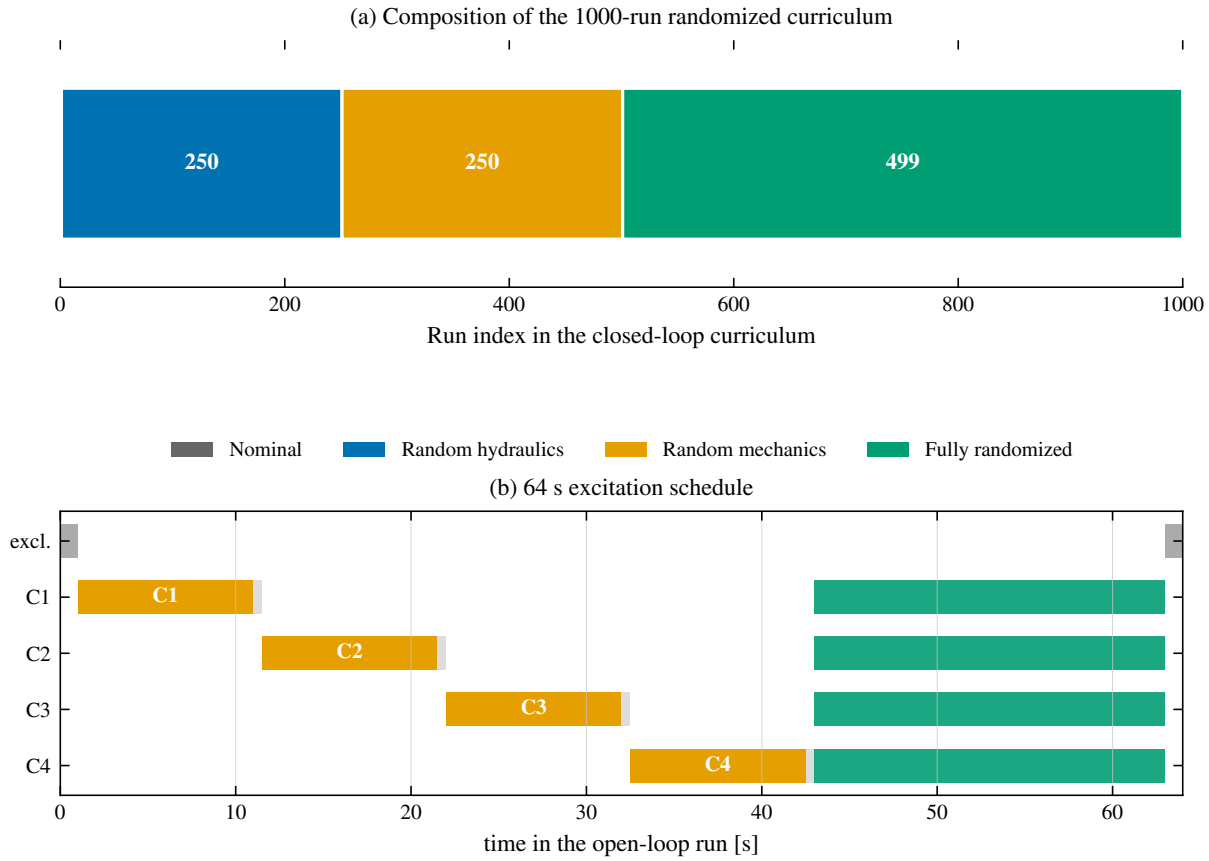


Figure 4.2: Design of the two collection campaigns. (a) Composition of the closed-loop curriculum: one nominal run, then 250 hydraulics-only, 250 mechanics-only and 499 fully randomized draws. (b) Timeline of one open-loop excitation run.

The seal-wear coordinate deserves comment because it is the one place where a single sampled number is deliberately allowed to move several model parameters together. A worn cylinder does not become leakier without also becoming slower, so $w_{\mathrm{seal},i}$ scales the piston clearance linearly, the valve leakage area with the cube of the same factor, and the spool time constant from 8 ms to about 13 ms at the top of the range. Sampling those three independently would have produced physically incoherent machines, some of them very leaky and yet perfectly fast.

4.2.3 The Compressibility Axis

The entrained-air and dead-volume coordinates randomize the chamber dead volume which spans 505 to 579 mL across the batch. This reaches the pressure dynamics through the continuity equation (3.4), which carries the chamber volume: a larger chamber is more compliant and slower, a smaller one stiffer and faster. The compressibility axis therefore enters the batch as capacitance variation.

4.2.4 Mechanical Parameters

The mechanical axes, listed in Table B.2, are applied to all fourteen rigid bodies of the mechanism: the six links, the four cylinder barrels, and the four pistons. For each body